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IFIP was founded in 1960 under the auspices of UNESCO, following the First World Computer Congress held in Paris the previous year. An umbrella organization for societies working in information processing, IFIP's aim is two-fold: to support information processing within its member countries and to encourage technology transfer to developing nations. As its mission statement clearly states,

> *IFIP's mission is to be the leading, truly international, apolitical organization which encourages and assists in the development, exploitation and application of information technology for the benefit of all people.*

IFIP is a non-profitmaking organization, run almost solely by 2500 volunteers. It operates through a number of technical committees, which organize events and publications. IFIP's events range from an international congress to local seminars, but the most important are:

- The IFIP World Computer Congress, held every second year;
- Open conferences;
- Working conferences.

The flagship event is the IFIP World Computer Congress, at which both invited and contributed papers are presented. Contributed papers are rigorously refereed and the rejection rate is high.

As with the Congress, participation in the open conferences is open to all and papers may be invited or submitted. Again, submitted papers are stringently refereed.

The working conferences are structured differently. They are usually run by a working group and attendance is small and by invitation only. Their purpose is to create an atmosphere conducive to innovation and development. Refereeing is also rigorous and papers are subjected to extensive group discussion.

Publications arising from IFIP events vary. The papers presented at the IFIP World Computer Congress and at open conferences are published as conference proceedings, while the results of the working conferences are often published as collections of selected and edited papers.

Any national society whose primary activity is about information processing may apply to become a full member of IFIP, although full membership is restricted to one society per country. Full members are entitled to vote at the annual General Assembly, National societies preferring a less committed involvement may apply for associate or corresponding membership. Associate members enjoy the same benefits as full members, but without voting rights. Corresponding members are not represented in IFIP bodies. Affiliated membership is open to non-national societies, and individual and honorary membership schemes are also offered.

More information about this series at http://www.springer.com/series/6102

José Abdelnour Nocera · Barbara Rita Barricelli
Arminda Lopes · Pedro Campos
Torkil Clemmensen (Eds.)

Human Work Interaction Design

Work Analysis and Interaction Design Methods for Pervasive and Smart Workplaces

4th IFIP 13.6 Working Conference, HWID 2015
London, UK, June 25–26, 2015
Revised Selected Papers

Springer

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Preface

Human Work Interaction Design (HWID) was established in September 2005 as the sixth working group (WG 13.6) of the IFIP Technical Committee 13 on Human–Computer Interaction (HCI). The scope of this group is the analysis and interaction design of a variety of complex work and life contexts found in different business and application domains. For this purpose, it is important to establish relationships between extensive empirical work domain studies and HCI design. WG 13.6 aims to provide the basis for an improved cross-disciplinary cooperation and mutual inspiration among researchers from the many disciplines that by nature are involved in deep analysis of a work domain. Complexity is hence a key notion in the activities of this working group, but it is not a priori defined or limited to any particular domains. WG 13.6 initiates and fosters new research initiatives and developments, as well as an increased awareness of HWID in the HCI curriculum.

This volume presents chapters developed from papers presented at the 4th HWID working conference held at the University of West London during June 25–26, 2015. The theme of this conference was on the integration of work analysis and interaction design methods for pervasive and smart workplaces.

Pervasive and smart technologies have pushed workplace configuration beyond linear logic and physical boundaries. As a result, workers' experience of and access to technology is increasingly pervasive, and their agency is constantly reconfigured. While in certain areas of work this is not new (e.g., technology mediation and decision support in air traffic control), more recent developments in other domains such as health care (e.g., augmented reality in computer-aided surgery) have raised challenging issues for HCI researchers and practitioners. The question now is how to improve the quality of workers' experience and outputs?

The chapters in this book focus on answering this question to support professionals, academia, national laboratories, and industry engaged in human work analysis and interaction design for the workplace. The first section provides an overview as well as instances of what could be classed as HWID methodologies. The second section offers different experiences on conceptualizing and researching the work environment in terms of how to sense and integrate its different dimensions into interaction design. The last section of the book presents chapters providing examples of HWID application in pervasive and smart workplaces across various domains such as aviation, education, product design, and archeology.

We hope this book becomes a resource for the type of discussion around HWID topics that took place at the working conference in London.

November 2015

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